

AGENT

design aid (optional)

DESIGN POLICY

choose treatments, replication,
and physical allocation D_r

EXPERIMENT

FACILITY + LATENT SITE θ

commit named units for a 210-day crop cycle
treatments remain immutable; occupied capacity cannot be released
timestamped interim measurements $\rightarrow$ terminal outcomes

inference aid (optional)

HISTORY READER

interpret visible history H_t
and recommend $\hat{x}_t$

BENCHMARK ONLY

EVALUATOR

no information returns to the agent

Exact outcomes

$\bar{\mathcal{R}}(r)$ recommendation regret

$\mathcal{R}_{\text{cum}}$ experimental opportunity cost

Common-history diagnostics

$b(H_t)$ risk left by the visible history

$g(H_t, \hat{x}_t)$ excess risk of the recommendation

access: recommendation + hidden truth/model

A closed-loop agent may implement design and reading in one model; fixed-history replay separates the roles experimentally.